Theodolite Survey

· It is used to measure horizontal and vertical angles accurately.

· It is also called as universal instrument.

Types of Theodolites

- **Based on Reversibility:**

Transit Theodolite: The telescope can be rotated by 180° (full rotation) in the vertical plane.

Non-Transit Theodolite: The telescope cannot be rotated by 180° (full rotation).

- **Based on System of Measurement:**

Vernier Theodolite:

· Least Count: 20"

Optical/Micrometre/Micro optic Theodolite:

· Least Count: 0.5" to 5"

· Readings are taken from a glass disc

Electronic Theodolite:

· Uses a bar-coded circular disc for readings

· Principle: Electro-optical scanning

· Least Count: 0.5" to 5"

Terms in Theodolite Operation

Transiting/Reversing/Plunging: Rotating the telescope about the horizontal axis in the vertical plane.

Swinging: Rotating the telescope in the horizontal plane about the vertical axis.

· Clockwise rotation: Right swing

· Anticlockwise rotation: Left swing

Face Left Observation: When the vertical circle is to the left of the telescope during an observation.

· Also known as telescope normal or bubble up

Face Right Observation: When the vertical circle is to the right of the telescope during an observation.

· Also known as telescope inverted or bubble down

Common Method: Face left, right swing

Changing Face: Switching from left face to right face or vice versa

Temporary Adjustments of Theodolite

Adjustments to be made in the field:

Centring: Using optical plummet, plumb bob, or shifting head

Levelling

Elimination of parallax:

· Focusing eyepiece

· Focusing objective

Permanent Adjustments of Theodolite

Plate Level Adjustment:

· When the vertical axis is truly vertical, the plate level bubble should be central

Cross Hair Ring Test:

· Horizontal hair should be perpendicular to the vertical axis of the instrument

Collimation in Azimuth Test:

· The line of collimation should be perpendicular to the horizontal axis or trunnion axis. (The horizontal axis is also known as the trunnion axis.)

Spire Test:

· To make the horizontal axis perpendicular to the vertical axis, or to make the horizontal axis truly horizontal.

· Done using a striding level.

Bubble Tube Adjustment:

· When the line of sight is horizontal, the altitude bubble should be central.

Vertical Arc Test:

· The vertical circle should read zero when the line of sight is perpendicular to the vertical axis.

· Measurement of Horizontal Angles with Theodolite:

Repetition Method

· Used to measure horizontal angles very accurately.

· Accuracy is better than what the vernier can provide.

· Minimum number of readings: 6 (3 left face and 3 right face)

Errors eliminated by repetition method:

a) Eccentricity error(main scale and vernier scale should be concentric): Eliminated by reading both verniers.

b) Collimation error (index error): Eliminated by both face observations.

c) Graduation error: Eliminated by reading at different parts of the main scale.

d) Sighting error: Eliminated by multiple observations.

Reiteration Method

· Used to measure angles around a centrally located instrument station.

· Useful in areas to be avoided, like water bodies, etc.

· Used to check areas to be irrigated.

· Sum of angles around the instrument station should equal 360° (known as "closing the horizon").